



Derivatives of Logarithmic Functions

Calculus Worksheet · Grade 11-12

Name: _____

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Learning Objectives

- Apply the derivative formulas for logarithmic and natural logarithmic functions
- Use the chain rule when differentiating composite logarithmic expressions
- Simplify derivatives using logarithm expansion properties

Find the derivative of each logarithmic function and simplify your answer completely.

1. Find the derivative of the function.

$$f(x) = \log_3(5x)$$

Answer: _____

2. Find the derivative of the function.

$$y = \log(2 + \sin x)$$

Answer: _____

3. Find the derivative of the function.

$$f(x) = \ln(x^2 + 1)$$

Answer: _____

4. Find the derivative of the function.

$$y = \ln(\sqrt{x})$$

Answer: _____

5. Find the derivative of the function.

$$f(x) = \log_2(x^3 - 4x)$$

Answer: _____

6. Find the derivative of the function.

$$y = \ln\left(\frac{x}{x+1}\right)$$

Answer: _____

7. Find the derivative of the function.

$$f(x) = \ln(\cos x)$$

Answer: _____



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8. Find the derivative of the function.

$$y = \log_5(4x^2 + 7)$$

Answer: _____

9. Find the derivative of the function.

$$f(x) = \ln(x^2 \cdot e^x)$$

Answer: _____

10. Find the derivative of the function.

$$y = \log_7(\tan x)$$

Answer: _____





Remind students that $\log x$ without a base implies base 10, and emphasize identifying u and du before applying the formula.

Solutions

1. Find the derivative of the function.

$$f(x) = \log_3(5x)$$

→ Identify u equals $5x$ and du equals 5 .

→ Apply the formula derivative of log base a of u equals du over u times $\ln a$.

→ Substitute to get 5 over $5x$ times $\ln 3$.

→ Cancel the 5 in numerator and denominator to get 1 over $x \ln 3$.

Answer: $f'(x) = \frac{1}{x \ln 3}$

2. Find the derivative of the function.

$$y = \log(2 + \sin x)$$

→ Identify u equals 2 plus sine x and du equals cosine x .

→ Recognize that log with no base means base 10, so use $\ln 10$.

→ Apply the logarithm derivative formula.

→ Write the simplified result as cosine x over the quantity 2 plus sine x times $\ln 10$.

Answer: $\frac{dy}{dx} = \frac{\cos x}{(2 + \sin x) \ln 10}$

3. Find the derivative of the function.

$$f(x) = \ln(x^2 + 1)$$

→ Identify u equals x squared plus 1 and du equals $2x$.

→ Apply the natural log derivative formula du over u .

→ Write the final answer as $2x$ over x squared plus 1 .

Answer: $f'(x) = \frac{2x}{x^2 + 1}$

4. Find the derivative of the function.

$$y = \ln(\sqrt{x})$$

→ Rewrite $\ln$ of square root of x as one half times $\ln x$ using the exponent expansion rule.

→ Differentiate one half times $\ln x$ to get one half times 1 over x .

→ Simplify to 1 over $2x$.

Answer: $\frac{dy}{dx} = \frac{1}{2x}$



5. Find the derivative of the function.

$$f(x) = \log_2(x^3 - 4x)$$

→ Identify u equals x cubed minus $4x$ and du equals $3x$ squared minus 4 .

→ Apply the log base a derivative formula with a equals 2 .

→ Write the result as $3x$ squared minus 4 all over the quantity x cubed minus $4x$ times $\ln 2$.

Answer:
$$f'(x) = \frac{3x^2 - 4}{(x^3 - 4x)\ln 2}$$

6. Find the derivative of the function.

$$y = \ln\left(\frac{x}{x+1}\right)$$

→ Expand using the quotient rule for logarithms to get $\ln x$ minus $\ln$ of x plus 1 .

→ Differentiate each term to get 1 over x minus 1 over x plus 1 .

→ Combine over a common denominator to get x plus 1 minus x over x times x plus 1 .

→ Simplify the numerator to 1 , giving 1 over x times x plus 1 .

Answer:
$$\frac{dy}{dx} = \frac{1}{x(x+1)}$$

7. Find the derivative of the function.

$$f(x) = \ln(\cos x)$$

→ Identify u equals cosine x and du equals negative sine x .

→ Apply the natural log derivative formula du over u .

→ Get negative sine x over cosine x .

→ Recognize this as negative tangent x .

Answer:
$$f'(x) = -\tan x$$

8. Find the derivative of the function.

$$y = \log_5(4x^2 + 7)$$

→ Identify u equals $4x$ squared plus 7 and du equals $8x$.

→ Apply the log base 5 derivative formula.

→ Write the simplified answer as $8x$ over the quantity $4x$ squared plus 7 times $\ln 5$.

Answer:
$$\frac{dy}{dx} = \frac{8x}{(4x^2 + 7)\ln 5}$$

9. Find the derivative of the function.

$$f(x) = \ln(x^2 \cdot e^x)$$

→ Expand the logarithm using the product rule to get $2 \ln x$ plus $\ln e$ to the x .

→ Simplify $\ln e$ to the x as just x .

→ Differentiate $2 \ln x$ plus x to get 2 over x plus 1 .

Answer:
$$f'(x) = \frac{2}{x} + 1$$



10. Find the derivative of the function.

$$y = \log_7(\tan x)$$

→ Identify u equals tangent x and du equals secant squared x .

→ Apply the log base 7 derivative formula.

→ Write the simplified result as secant squared x over the quantity tangent x times $\ln 7$.

Answer: $\frac{dy}{dx} = \frac{\sec^2 x}{\tan x \cdot \ln 7}$

